Chapter 6 Study Sheet

Relationships Within Triangles — everything you need on one page.

6.1 Perpendicular and Angle Bisectors

- Perpendicular Bisector Theorem: a point on the perpendicular bisector is EQUIDISTANT from the endpoints — and conversely.
- Angle Bisector Theorem: a point on an angle bisector is EQUIDISTANT from the two sides — and conversely, for a point in the interior.
- Distance from a point to a line means the length of the PERPENDICULAR segment — nothing else counts.
- To write a perpendicular bisector: midpoint, opposite reciprocal of the slope, point-slope form. Check the midpoint satisfies it.

6.2 Bisectors of Triangles

- Concurrent lines pass through one common point, called the point of concurrency.
- Circumcenter Theorem: the three PERPENDICULAR BISECTORS meet at a point equidistant from the three VERTICES.
- Incenter Theorem: the three ANGLE BISECTORS meet at a point equidistant from the three SIDES.
- Circumcenter: inside (acute) · at the midpoint of the hypotenuse (right) · outside (obtuse). The incenter is ALWAYS inside.
- On coordinates, use a vertical and a horizontal side — their perpendicular bisectors are y = ___ and x = ___.

6.3 Medians and Altitudes

- A MEDIAN joins a vertex to the MIDPOINT of the opposite side; the three medians meet at the CENTROID, always inside.
- Centroid Theorem: the centroid is 2/3 of the way from each vertex, so vertex-to-centroid = 2 × centroid-to-midpoint.
- Centroid from coordinates: average the three x's and average the three y's.
- An ALTITUDE drops PERPENDICULARLY from a vertex to the opposite side's line; the three meet at the ORTHOCENTER — inside (acute) · at the right-angle vertex (right) · outside (obtuse).

6.4 The Triangle Midsegment Theorem

- A midsegment joins the midpoints of two sides. Every triangle has three.
- Each midsegment is PARALLEL to the third side and HALF its length.
- So the midsegment triangle has half the perimeter of the original.
- On coordinates: find the two midpoints, compare slopes (parallel) and lengths (half).

6.5 Indirect Proof and Inequalities in Triangles

- Bigger side ⟷ bigger angle. Order the sides and you have ordered the angles opposite them.
- Triangle Inequality Theorem: any two sides must sum to MORE than the third. Equal means collinear — no triangle.
- Third-side range: $|a - b| < x < a + b$.
- Indirect proof: assume the opposite, reason to a contradiction, conclude the original statement is true.